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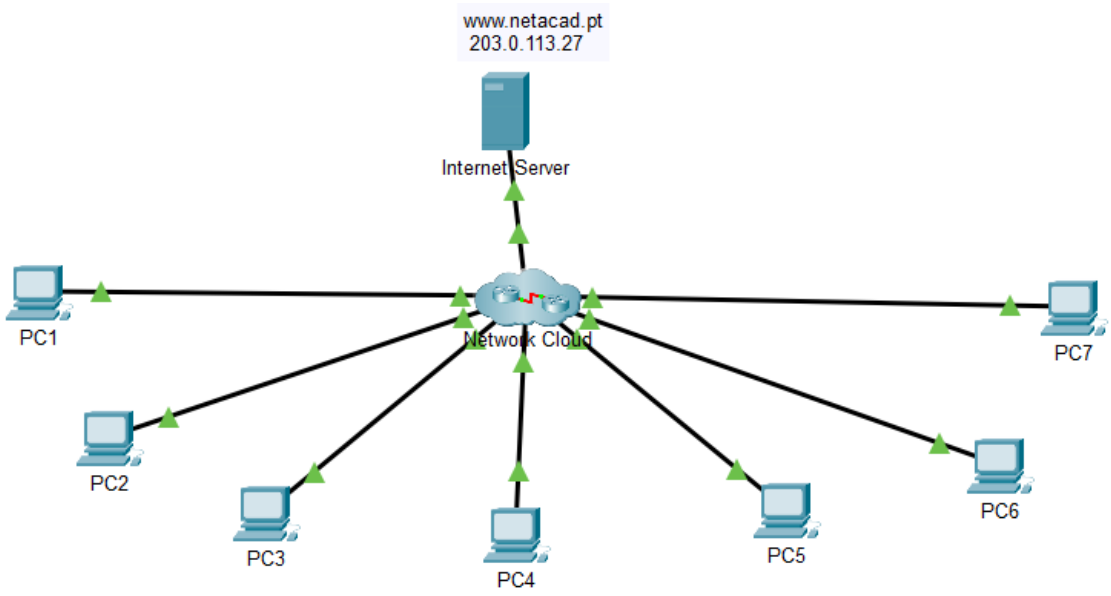
Labo PT 12.6.2



1) Cas de PC1 (Réseau Branch 1)

Méthodologie : Sur PC1, je tente de ping le serveur (**203.0.113.27**). Comme le ping échoue, j'utilise la commande tracert pour localiser le point de rupture, puis je vérifie les tables du routeur où le paquet s'arrête.

1) Problème de Routage (ISP)

Tests effectués :

- **Ping 203.0.113.27 :** Échec .
- **Tracert 203.0.113.27 :** Le paquet atteint **192.168.0.1** (Hub) mais s'arrête ensuite (* * *).
- **Commande sur ISP :** show ip route. Je constate qu'il n'y a aucune route pour le réseau **192.168.1.0/24** .

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>tracert 203.0.113.27

Tracing route to 203.0.113.27 over a maximum of 30 hops:

  1   0 ms      1 ms      0 ms      192.168.1.1
  2   2 ms      1 ms      9 ms      192.168.0.1
  3   *         *         *         Request timed out.
  4   *         *         *         Request timed out.
  5   *         *         *         Request timed out.
  6   *         *         *         Request timed out.
```

```
ISP#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.0/30 is directly connected, GigabitEthernet0/0/0
L       192.0.2.2/32 is directly connected, GigabitEthernet0/0/0
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/24 is directly connected, GigabitEthernet0/0/1
L       203.0.113.1/32 is directly connected, GigabitEthernet0/0/1
     209.165.201.0/27 is subnetted, 1 subnets
S       209.165.201.128/27 [1/0] via 192.0.2.1

ISP#
```

Problème : Le routeur **ISP** ne connaît pas le chemin de retour pour répondre aux PC de la branche 1.

Solution : Ajouter une route statique sur l'ISP pointant vers l'interface du Hub.

Code: ISP(config)# ip route 192.168.1.0 255.255.255.0 192.0.2.1

2) Problème de NAT (Hub)

Tests effectués :

- **Commande sur Hub :** show ip nat translations.
- **Observation :** Le tableau est vide alors que des requêtes sortantes sont envoyées. Cela indique que le trafic n'est pas traduit.

```
Hub#
Hub#sh ip nat translations
Hub#conf t
```

Problème : Le NAT n'est pas activé sur les interfaces du Hub. Les adresses privées ne sont pas converties en adresses publiques, ce qui est indispensable pour circuler sur Internet.

Solution : Configurer les interfaces inside et outside, créer une ACL et activer la surcharge (overload).

Code :

```
Hub(config)# interface GigabitEthernet0/0/0
Hub(config-if)# ip nat outside
Hub(config)# interface Serial0/1/0
Hub(config-if)# ip nat inside
Hub(config)# access-list 1 permit 192.168.0.0 0.0.255.255
Hub(config)# ip nat inside source list 1 interface GigabitEthernet0/0/0 overload
```

Test de validation final : Après avoir appliqué ces correctifs, je relance le test sur **PC1** :

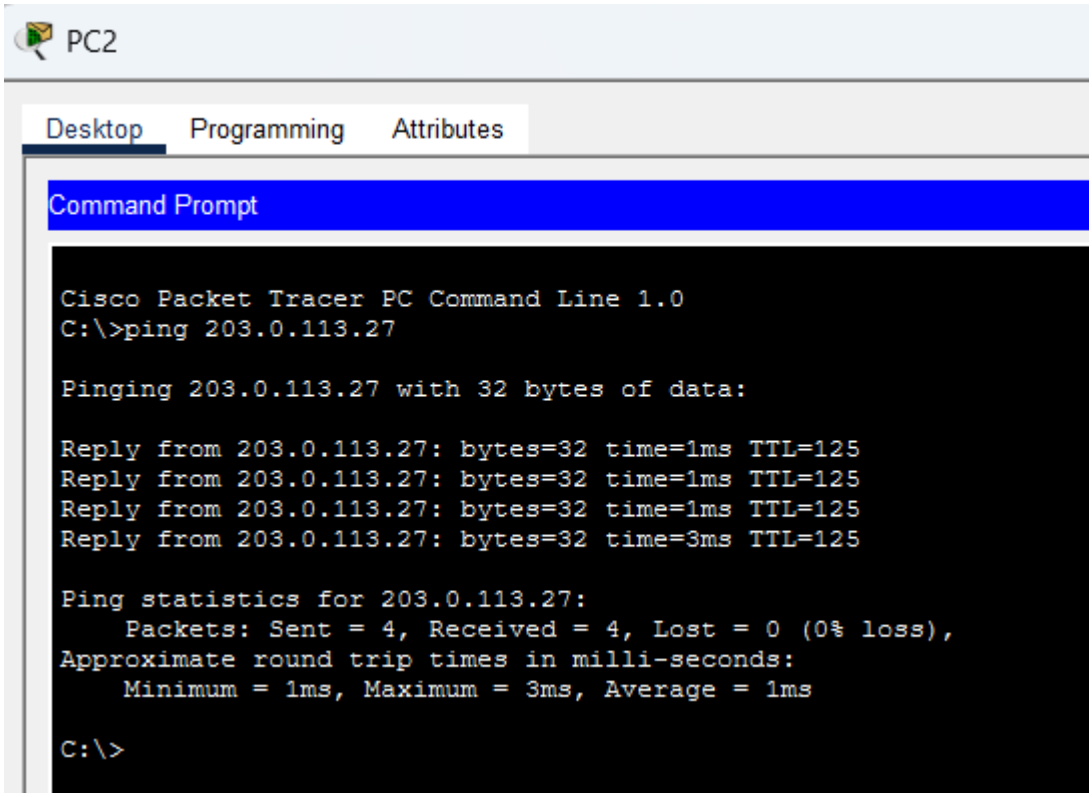
- ping 203.0.113.27 -> **Succès**

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Reply from 203.0.113.27: bytes=32 time=20ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125
Reply from 203.0.113.27: bytes=32 time=9ms TTL=125
Reply from 203.0.113.27: bytes=32 time=11ms TTL=125

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 10ms

C:\>
```

2) Cas de PC2 (Vérification de connectivité)

Test de validation : test sur **PC2** :

ping 203.0.113.27 -> **Succès**



3. Cas de PC3 (Réseau Factory)

Méthodologie : Le ping vers le serveur échoue. J'analyse la configuration OSPF du routeur Factory pour vérifier si le réseau local est bien exporté vers le Hub.

- **Tests effectués :**
  - **Ping 203.0.113.27 :** Échec.
  - **Show run sur Factory :** Je constate que l'interface LAN (G0/0/1) est dans le processus ospf 10

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
router ospf 10
 log-adjacency-changes
 network 192.168.3.0 0.0.0.255 area 0
 network 192.168.0.12 0.0.0.3 area 0
```

- **Problème :** le router Hub ne connait pas le chemin pour atteindre le réseau 192.168.4.0
- **Solution :** annoncer le réseau dans ospf 10 sur factory
- **Code :**

Factory(config)# router ospf 10
Factory(config-router)# network 192.168.4.0 0.0.0.255 area 0

```
router ospf 10
 log-adjacency-changes
 network 192.168.3.0 0.0.0.255 area 0
 network 192.168.0.12 0.0.0.3 area 0
 network 192.168.4.0 0.0.0.255 area 0
```

Test de validation final : Après avoir appliqué ces correctifs, je relance le test sur **PC3** :

ping 203.0.113.27 -> **Succès**

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Reply from 203.0.113.27: bytes=32 time=12ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 3ms

C:\>
```

---

4) Cas de PC4 (Vérification de connectivité)

Test de validation : test sur **PC4** :

ping 203.0.113.27 -> **Succès**

PC4

Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Reply from 203.0.113.27: bytes=32 time=1ms TTL=125
Reply from 203.0.113.27: bytes=32 time=8ms TTL=125
Reply from 203.0.113.27: bytes=32 time=6ms TTL=125
Reply from 203.0.113.27: bytes=32 time=7ms TTL=125

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 8ms, Average = 5ms

C:\>
```

5) Cas de PC5

Méthodologie : Les liaisons physiques sont rétablies (vertes), mais le PC ne communique toujours pas. Je vérifie les paramètres IP de l'hôte.

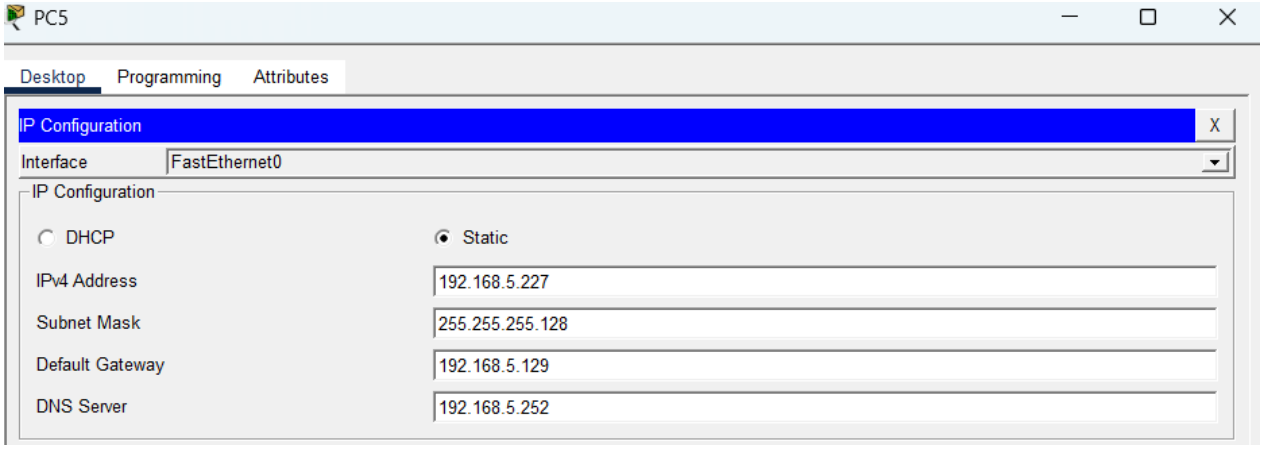
- **Tests effectués :**
  - **Ping 203.0.113.27**
  - **ipconfig sur PC5 :** L'adresse IP est 192.168.5.227, mais la **Default Gateway** est 0.0.0.0.

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 203.0.113.27
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::200:CFF:FE1B:50E7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.5.227
   Subnet Mask.....................: 255.255.255.128
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ipconfig
```

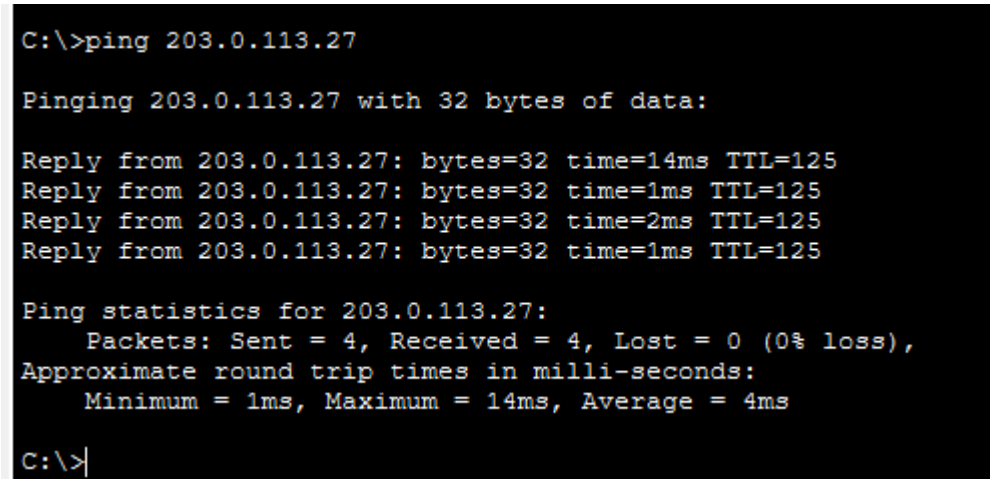
- **Problème :** Absence de passerelle par défaut sur le poste client. Le PC ne sait pas envoyer ses paquets vers le routeur pour sortir du réseau.
- **Solution :** Configurer la passerelle 192.168.5.129

- **Action :** Saisie de 192.168.5.129 dans les paramètres IPv4 de PC5.



Test de validation final : Après avoir ajouter la passerelle, je relance le test sur **PC5** :

ping 203.0.113.27 -> **Succès**



```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Reply from 203.0.113.27: bytes=32 time=14ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125
Reply from 203.0.113.27: bytes=32 time=2ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 14ms, Average = 4ms

C:\>
```

6) Cas de PC6 (Réseau Branch-2)

Méthodologie : Le ping affiche "Destination host unreachable". Ce message indique que le routeur local ne sait pas acheminer le paquet.

- **Tests effectués :**
  - **Ping 203.0.113.27 :** Hôte de destination injoignable
  - **sh ip int br sur Branch-2 :** L'interface WAN (S0/1/0) a l'IP 192.168.0.17.

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Reply from 192.168.2.33: Destination host unreachable.
Reply from 192.168.2.33: Destination host unreachable.
Reply from 192.168.2.33: Destination host unreachable.
Reply from 192.168.2.33: Destination host unreachable.

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

```
Branch-2#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES manual up                    up
GigabitEthernet0/0/0.1 192.168.2.1     YES manual up                    up
GigabitEthernet0/0/0.32192.168.2.33    YES manual up                    up
GigabitEthernet0/0/0.64192.168.2.65    YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
Serial0/1/0            192.168.0.17    YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Branch-2#conf t
```

- **Problème :** une mauvaise ip a été attribué à s0/1/0 de Brunch-2

- **Solution :** configurer 192.168.0.6/30.

- **Code :**

```
Branch-2(config)# interface s0/1/0
Branch-2(config-if)# ip address 192.168.0.6 255.255.255.252
```

```
Branch-2#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES manual up                    up
GigabitEthernet0/0/0.1 192.168.2.1     YES manual up                    up
GigabitEthernet0/0/0.32192.168.2.33    YES manual up                    up
GigabitEthernet0/0/0.64192.168.2.65    YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
Serial0/1/0            192.168.0.6     YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Branch-2#
```

Test de validation final : Après avoir configurer 192.168.0.6, je relance le test sur **PC6** :

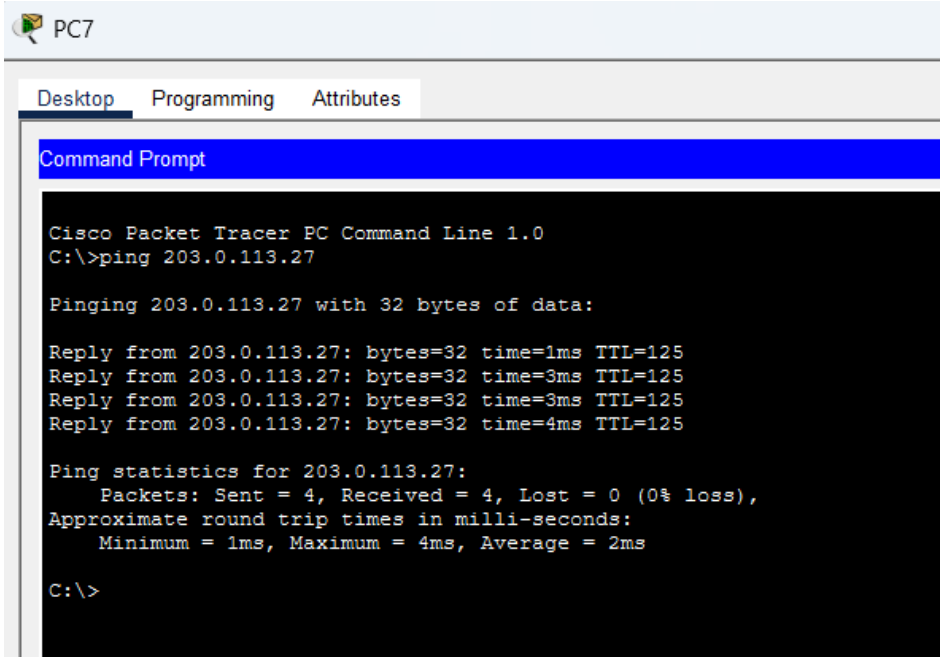
ping 203.0.113.27 -> **Succès**

```
C:\>ping 203.0.113.27

Pinging 203.0.113.27 with 32 bytes of data:

Reply from 203.0.113.27: bytes=32 time=13ms TTL=125
Reply from 203.0.113.27: bytes=32 time=8ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125
Reply from 203.0.113.27: bytes=32 time=1ms TTL=125

Ping statistics for 203.0.113.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 13ms, Average = 5ms

C:\>
```

7) Cas de PC7 (Vérification de connectivité)

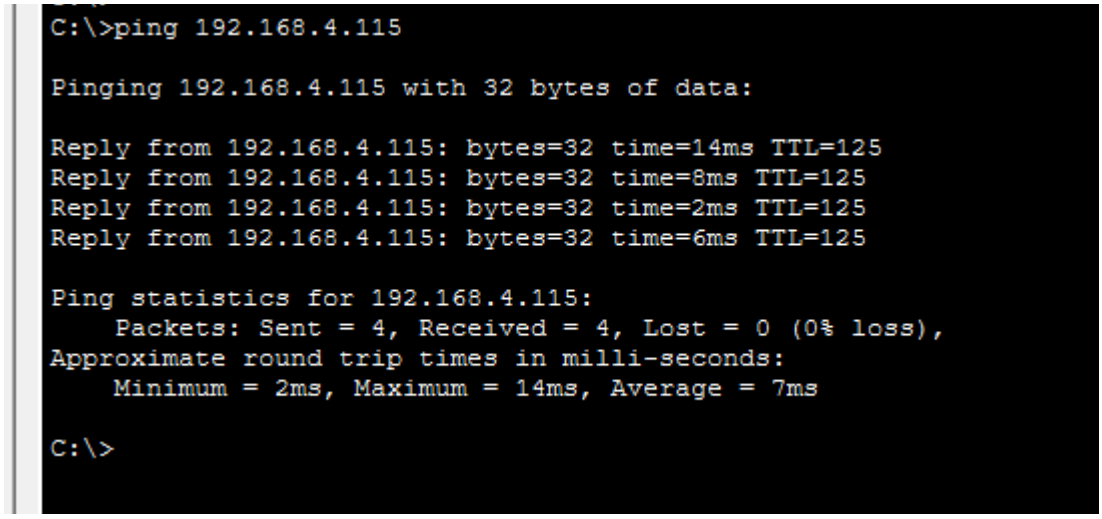
Test de validation : test sur **PC7** :

ping 203.0.113.27 -> **Succès**



8) ping depuis PC1 vers PC3

ping 192.168.4.115 -> **Succès**



9) ping depuis PC4 vers PC6

ping 192.168.2.48 -> **Succès**

```
C:\>ping 192.168.2.48

Pinging 192.168.2.48 with 32 bytes of data:

Reply from 192.168.2.48: bytes=32 time=2ms TTL=125
Reply from 192.168.2.48: bytes=32 time=6ms TTL=125
Reply from 192.168.2.48: bytes=32 time=8ms TTL=125
Reply from 192.168.2.48: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.2.48:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 6ms

C:\>
```

10) ping depuis PC7 vers PC2

ping 192.168.3.50 -> **Succès**

```
C:\>ping 192.168.3.50

Pinging 192.168.3.50 with 32 bytes of data:

Reply from 192.168.3.50: bytes=32 time=8ms TTL=125
Reply from 192.168.3.50: bytes=32 time=2ms TTL=125
Reply from 192.168.3.50: bytes=32 time=13ms TTL=125
Reply from 192.168.3.50: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 6ms

C:\>
```